Lee Hyoseok

hyoseok@kaist.ac.kr
Website | Scholar | Github

RESEARCH INTEREST

Exploring the intersection of computer vision & graphics, and machine learning, but not limited to.

Generative Models, Computational Imaging, and 3D Reconstruction

EDUCATION

Korea Advanced Institute of Science and Technology, KAIST — Sep. 2025 – Jan. 2026
Ph.D candidate, School of Computing (Advisor: Prof. Tae-Hyun Oh) — Daejeon, Korea

Pohang University of Science and Technology, POSTECH — Sep. 2023 – Aug.2025
M.S., Artificial Intelligence (Advisor: Prof. Tae-Hyun Oh) — Pohang, Korea

Kookmin University, KMU — Mar. 2019 – Aug. 2023
Bachelor, Major: Big Data Management Statistics & Minor: Computer Science — Seoul, Korea

EXPERIENCE

Undergraduate Research Internship — Jan. 2023 – May. 2023
Computer Graphics Lab (Advisor: Prof.Seung-Hwan Baek) — POSTECH, Korea

- Conducted research on event-based computational imaging

Student Representative — Aug. 2022 – Mar. 2023
Google Developer Student Club Lead — Kookmin University, Korea

- Worked on student leadership and AI mentoring

Undergraduate Research Internship — Jun. 2021 – Mar. 2023
Visual Computing Lab(Advisor: Prof. Junho Kim) — Kookmin University, Korea

- Conducted research on NeRF-based 3D generative models

AWARD & HONOR

Best Excellence Prize, Electronics Times ICT Paper Awards — Dec. 2025
"Measurement-Consistent Langevin Corrector: A Remedy for Latent Diffusion Inverse Solvers".

Winner, Qualcomm Innovation Fellowship Korea 2025 — Nov. 2025
"Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior".

Excellence Prize, Electronics Times ICT Paper Awards — Nov. 2024
"FPGS: Feed-Forward Semantic-aware Photorealistic Style Transfer of Large-Scale Gaussian Splatting".

Outstanding Poster Presentation Awards for IPIU 2024 — Feb. 2024
"Bootstrapping Multi-View Features via Bridging the Gap between Linearity of Rendering and Non-linearity of 2D Feature Space".

TEACHING & ACADEMIC SERVICE

Teaching Assistant

- "Resource-Efficient AI" lecture on Dept. of CS, KAIST, 2025 Fall
- "Human-centric Social AI System Design" lecture on Dept. of EE, POSTECH, 2024 Fall
- "Image Processing" lecture on Dept. of EE, POSTECH, 2024 Spring

PUBLICATION

Journal

[J1] G. Kim, K. Youwang, **Lee Hyoseok**, T.-H. Oh, “FPGS: Feed-Forward Semantic-aware Photorealistic Style Transfer of Large-Scale Gaussian Splatting” *IJCV*.
(Excellence Prize, Electronics Times ICT Paper Awards 2024)

Conference

[C7] S. Lim, **Lee Hyoseok**, J. Park, T.-H. Oh, “CLAY: Conditional Visual Similarity Modulation in Vision-Language Embedding Space", *CVPR*, 2026.

[C6] K. Youwang, **Lee Hyoseok**, P. Subin, G. Pons-Moll, T.-H. Oh, “ELITE: Efficient Gaussian Head Avatar from a Monocular Video via Learned Initialization and TEst-time Generative Adaptation", *CVPR*, 2026.

[C5] **Lee Hyoseok**, S. Lim, E. Cha, T.-H. Oh, “Measurement-Consistent Langevin Corrector: A Remedy for Latent Diffusion Inverse Solvers", *arXiv*, 2025.
(Best Excellence Prize, Electronics Times ICT Paper Awards 2025)

[C4] K. Youwang, **Lee Hyoseok**, G. Pons-Moll, T.-H. Oh, “Dress-up: Generating Animatable Clothed 3D Humans via Latent Modeling of 3D Gaussian Texture Maps" *ICCV Workshop on Computer Vision for Fashion, Art, and Design*, 2025.
(Oral Presentation)

[C3] S. Lim, **Lee Hyoseok**, T.-H. Oh, “Conditioned Image-to-Image Retrieval via Concept-based Visual Projections in Vision-Language Model" *CVPR Workshop on Computer Vision in the Wild*, 2025.

[C2] B.-K. Kwon, Q. Dai, **Lee Hyoseok**, C. Luo, T.-H. Oh, “JointDiT: Enhancing RGB-Depth Joint Modeling with Diffusion Transformers" *ICCV*, 2025.

[C1] **Lee Hyoseok**, K. S. Kim, B.-K. Kwon, T.-H. Oh, “Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior" *AAAI*, 2025.
(Winner of the Qualcomm Innovation Fellowship Korea, 2025)

Domestic

[D1] **Lee Hyoseok**, K. Jun-Seong, K . Ji-Yeon, T.-H. Oh, “Bootstrapping Multi-View Features via Bridging the Gap between Linearity of Rendering and Non-linearity of 2D Feature Space" *36th Workshop on Image Processing and Image Understanding*, 2024.
(Outstanding Poster Presentation Award)

PATENT

Method, computing device and computer program for generating depth map based on point cloud using artificial intelligence	KR10-2024-0170513

REFERENCE

Prof. Tae-Hyun Oh, Professor, KAIST
Relationship: M.S. advisor
E-mail: thoh.kaist.ac.kr@gmail.com

Prof. Seung-Hwan Baek, Professor, POSTECH
Relationship: undergraduate advisor
E-mail: shwbaek@postech.ac.kr

Prof. Junho Kim, Professor, Kookmin Univ.
Relationship: undergraduate advisor
E-mail: junho@kookmin.ac.kr